
DSC 40A - Homework 4

due Tuesday, July 14th at 11:59 PM PT

Homeworks are due to Gradescope by 11:59PM on the due date.

You may request an extension for this assignment by submitting [the form](#).

Homework will be evaluated not only on the correctness of your answers, but on your ability to present your ideas clearly and logically. You should **always explain and justify** your conclusions, using sound reasoning. Your goal should be to convince the reader of your assertions. If a question does not require explanation, it will be explicitly stated.

Homeworks should be written up and turned in by each student individually. You may talk to other students in the class about the problems and discuss solution strategies, but you should not share any written communication and you should not check answers with classmates. You can tell someone how to do a homework problem, but you cannot show them how to do it. **Only handwritten solutions will be accepted (use of tablets is permitted). Do not typeset your homework (using L^AT_EX or any other software).** Staff will not grade typed submissions.

For each problem you submit, you should **cite your sources** by including a list of names of other students with whom you discussed the problem. Instructors do not need to be cited.

This homework will be graded out of **24 points**. The point value of each problem or sub-problem is indicated by the number of avocados shown.

Problem 1. Reflection and Feedback Form



Make sure to fill out this Reflection and Feedback Form, linked [here](#), for two points on this homework! This form is primarily for your benefit; research shows that reflecting and summarizing knowledge helps you understand and remember it.

Problem 2. Same, but Different

In Lecture 3, we were introduced to one of many formulas for the optimal slope, w_1^* , and optimal intercept, w_0^* , for the simple linear regression model $H(x) = w_0 + w_1x$ when using squared loss:

$$w_1^* = r \frac{\sigma_y}{\sigma_x} \quad w_0^* = \bar{y} - w_1^* \bar{x}$$

Then, in Lecture 4, we revisited the simple linear regression model in terms of linear algebra. When $X \in \mathbb{R}^{n \times 2}$ is the design matrix, $\vec{y} \in \mathbb{R}^n$ is the observation vector, and $\vec{w} \in \mathbb{R}^2$ is the parameter vector, we found that the optimal parameter vector $\vec{w}^*$ is one that satisfies the normal equations:

$$X^T X \vec{w} = X^T \vec{y}$$

When $X^T X$ is invertible, $\vec{w}^*$ can be expressed as:

$$\vec{w}^* = (X^T X)^{-1} X^T \vec{y}$$

In this problem, we will prove that both of these formulations are equivalent, for any dataset $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. Specifically, we'll show that the vector $\vec{w}^* = (X^T X)^{-1} X^T \vec{y}$ has two components, the first of which is w_0^* and the second of which is w_1^* . (To do this, we'll need to assume that $(X^T X)^{-1}$ is invertible.)

Note that on first glance, it looks like this problem is quite long, since it has seven subparts. However, the subparts are meant to guide you through the proof. (The problem would take much longer if we just said "prove it!")

- a) 🥑🥑 Express the vector $X^T \vec{y}$ using constants and/or summations involving x_i and/or y_i . Make sure that your answer has the correct dimensions.
- b) 🥑🥑 Express the matrix $X^T X$ using constants and/or summations involving x_i and/or y_i . Make sure that your answer has the correct dimensions.
- c) 🥑🥑 Recall, if $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a 2×2 matrix, then the inverse of M is given by:

$$M^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Express the matrix $(X^T X)^{-1}$ using constants and/or summations involving x_i and/or y_i .

Optional: The rest of this problem is optional, meaning we will not grade or look at any responses to parts d-g. We encourage you to do it as practice for the upcoming exam.

- d) 🥑🥑 At this point, the expressions you have for $(X^T X)^{-1}$ and $X^T \vec{y}$ likely involve many summation notations and look... complicated. Let's take a step back and simplify things before we proceed.

Prove that:

$$\sum_{i=1}^n x_i^2 = n\sigma_x^2 + n\bar{x}^2$$

Hint: Start with the definition of σ_x^2 and expand the sum.

e) 🥑🥑 Using your work in (c) and (d), prove that:

$$(X^T X)^{-1} = \frac{1}{n\sigma_x^2} \begin{bmatrix} \sigma_x^2 + \bar{x}^2 & -\bar{x} \\ -\bar{x} & 1 \end{bmatrix}$$

f) 🥑🥑 $(X^T X)^{-1}$ is about as simplified as it can be for now. But, before we multiply $(X^T X)^{-1}$ and $X^T y$, we should deal with the fact that at least one of the components in the vector $X^T \vec{y}$ still involves a summation.

Prove that:

$$\sum_{i=1}^n x_i y_i = nr\sigma_x\sigma_y + n\bar{x}\bar{y}$$

g) 🥑🥑🥑🥑 Now, put it all together. That is, prove that:

$$(X^T X)^{-1} X^T \vec{y} = \begin{bmatrix} \bar{y} - r \frac{\sigma_y}{\sigma_x} \bar{x} \\ r \frac{\sigma_y}{\sigma_x} \end{bmatrix}$$

Note that the second component of the vector above is $w_1^* = r \frac{\sigma_y}{\sigma_x}$ and the first component of the vector above is $w_0^* = \bar{y} - r \frac{\sigma_y}{\sigma_x} \bar{x} = \bar{y} - w_1^* \bar{x}$, as we first saw in Lecture 5! This concludes our proof that both formulations of the optimal parameters of the simple linear regression model are equivalent.

Problem 3. Real Estate

You are given a data set containing information on recently sold houses in San Diego, including

- square footage
- number of bedrooms
- number of bathrooms
- year the house was built
- asking price, or how much the house was originally listed for, before negotiations
- sale price, or how much the house actually sold for, after negotiations

The table below shows the first few rows of the data set. Note that since you don't have the full data set, you cannot answer the questions that follow based on calculations; you must answer conceptually.

House	Square Feet	Bedrooms	Bathrooms	Year	Asking Price	Sale Price
1	1247	3	3	2005	500,000	494,000
2	1670	3	2	1927	1,000,000	985,000
3	716	1	1	1993	335,000	333,850
4	1600	4	2	1962	830,000	815,000
5	2635	4	3	1993	1,250,000	1,250,000
⋮	⋮	⋮	⋮	⋮	⋮	⋮

- a) 🥑🥑 First, suppose we fit a multiple linear regression model to predict the sale price of a house given all five of the other variables. Which feature would you expect to have the largest magnitude weight? Why? (Note that the weight of a feature is the value of w^* for that feature.)

Then, suppose we standardize each variable separately, i.e. we convert each variable to standard units. (Recall, to convert a variable to standard units, we replace each value x_i with $\frac{x_i - \bar{x}}{\sigma_x}$.) Suppose we fit another multiple linear regression model to predict the sale price of a house given all five of the other standardized variables. Now, which feature would you expect to have the largest magnitude weight? Why?

- b) 🥑🥑 Suppose we fit a multiple linear regression model to predict the sale price of a house given all five of the other variables in their original, unstandardized form. Suppose the weight for the Year feature is α .

Now, suppose we replace Year with a new feature, Age, which is 0 if the house was built in 2024, 1 if the house was built in 2023, 2 if the house was built in 2022, and so on. If we fit a new multiple linear regression model on all five variables, but using Age instead of Year, what will the weight for the Age feature be, in terms of α ?

- c) 🥑🥑 Now, suppose we fit a multiple linear regression model to predict the sale price of a house given all five of the other variables, plus a new sixth variable named Rooms, which is the total number of bedrooms and bathrooms in the house. Will our new regression model with an added sixth feature make better predictions than the models we fit in (a) or (b)?

- d) 🥑🥑🥑 Now, suppose we fit two multiple linear regression models to predict the sale price of a house. The first uses the features “Square feet” and “Bedrooms”:

$$H(\gamma_0, \gamma_1, \gamma_2) = \gamma_0 + \gamma_1 x^{(1)} + \gamma_2 x^{(2)}$$

The second model uses the features “Square feet” and “Bedrooms” and a new sixth feature named “Length of street name”, which is the number of letters in the name of the street that the house is on:

$$H'(\lambda_0, \lambda_1, \lambda_2, \lambda_3) = \lambda_0 + \lambda_1 x^{(1)} + \lambda_2 x^{(2)} + \lambda_3 x^{(6)}$$

Prove that $\text{MSE}(H') \leq \text{MSE}(H)$ always.

Problem 4. The Behavior of Gradient Descent

Sometimes in optimization, it is difficult or impossible to find a closed-form solution for the objective function, such as the formulae for w_0 and w_1 in least squares regression. In such cases, one can use an iterative method such as gradient descent to find the optimal solution (or get close to it).

Recall the iterative step/update rule in gradient descent:

$$h_i = h_{i-1} - \alpha \cdot \frac{dR}{dh}(h_{i-1})$$

where α is a chosen learning rate.

- a) 🥑🥑🥑🥑 In lecture, you may have encountered the phrase, “given an appropriate step size, α ” or “small enough α ”. Let’s experiment with what this might mean. Suppose one chooses a starting prediction of $h_0 = 10$. Run two iterations of gradient descent with $\alpha = 1$ and $\alpha = 0.5$ on the function $f(x) = x^2$. Draw a graph depicting the results of the two gradient descent algorithms and explain which α is a better choice of a learning rate over the other.

Optional: The rest of this problem is optional, meaning we will not grade or look at any responses to part b. We encourage you to do it as practice for the upcoming exam.

- b) 🥑🥑🥑🥑 Find the optimal solution using calculus methods (i.e., find the critical points) for the following function,

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x_1, x_2) = 3x_1^2 + 0.5x_2^2.$$

Suppose one runs gradient descent on f with a small enough α . Write the update rule for f . Explain whether x_1 or x_2 reaches their respective minimum value at a faster rate than the other.